Fiscal Year:	FY 2026	**Task Last Updated:**	FY 12/12/2025
PI Name:	Norcross, Jason M.S.		
Project Title:	Validation of Fitness for Duty Standards Using Pre- and Post-Flight Capsule Egress and Suited Functional Performance Tasks in Simulated Reduced Gravity		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **EVA**:Risk of Mission Impacting Injury and Compromised Performance and Long-Term Health Effects due to EVA Operations (2) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058-3711	**Congressional District:**	36
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	01/30/2019	**End Date:**	12/01/2027
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/1/2027 per HHC element/JSC (Ed., 12/14/20)		
Key Personnel Changes/Previous PI:	There have been no significant changes to the study team over the past year.		
COI Name (Institution):	Marshall-Goebel, Karina Ph.D. (NASA Johnson Space Center) Young, Millennia Ph.D. (NASA Johnson Space Center) Schlotman, Taylor (KBR) Peters, Brian (KBR) Cox, Lauren (JES Tech) Strock, Nicole Ph.D. (KBR)		
Grant/Contract No.:	Internal Project		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	December 2025 Update: The Principal Investigator (PI) notes that the Pilot Egress Fitness protocol has expanded to include private astronaut missions (Ed., 1/8/26). Rigorous adherence to available inflight countermeasures has effectively mitigated losses or maintained muscle strength and aerobic capacity in some returning long-duration International Space Station (ISS) crewmembers; however, all astronauts demonstrate significant decrements in functional performance upon return to a gravity environment. These losses in functional performance can be largely attributed to neurovestibular / sensorimotor deficits that can take days or weeks from which to recover and for which there is no current operational countermeasure. Although these losses are tolerable for current land-based returns to Earth, where ground personnel can quickly support the crew at the landing site, this will not be the case for future off-nominal water-based Orion landings or for nominal Mars surface landings, both of which will require crewmembers to be capable of egressing their landing vehicle unassisted. Quantification of astronauts' post-landing functional capacity including ability to perform an unassisted capsule egress and critical planetary extravehicular activity (EVA) tasks is necessary to design concepts of operation for Moon and Mars exploration mission systems and ultimately to promote exploration mission success. These results can then be reviewed in combination with other pre-flight, in-flight, and post-landing measures and determinants of health and performance (e.g., sleep, nutrition, exercise) to help develop and select necessary countermeasures capable of protecting all crewmembers or to identify characteristics (both behavioral and inherent) that might allow for selection of crew dependent on mission objectives. Data collected in this proposal will provide unique data on unassisted capsule egress while either shirt-sleeved or wearing an unpressurized launch, entry, abort (LEA) suit in Earth's gravity and on EVA-relevant functional task performance by testing astronauts shortly after return to Earth while suited and pressurized in a simulated reduced gravity analog. The research product will be a temporal profile of unassisted capsule egress and planetary EVA task performance pre-flight and at multiple post-landing intervals, the timing of which will be determined based on post-landing logistics and coordination with other investigations. Data will be collected for spaceflight missions ranging from 2 months, 6 months, and up to 1 year in duration. Results of the proposed study will be used in combination with subsequent definition and design of exploration mission systems and operations concepts to define data-based Fitness for Duty standards.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The core focus of this study is to facilitate safe exploration of Mars and return back to Earth, which albeit has limited direct impact to the people on Earth, but to the extent that space exploration is for the good of all humanity, this study will facilitate more successful missions to Mars.
Task Progress:	NASA Institutional Review Board (IRB) Approval has been completed and maintained for the pilot study (JSC eIRB STUDY00000242) and CIPHER Egress Fitness (JSC eIRB STUDY00000169) and test readiness review (TRR) documentation has been completed and maintained for both studies. [Ed. Note: CIPHER is the short title for a set of 14 studies sponsored by NASA and international partner agencies (https://). CIPHER stands for "Complement of Integrated Protocols for Human Exploration Research".] The Pilot Egress Fitness IRB protocol was updated and expanded in 2025 to include a unique opportunity to collect data on the SpaceX Fram2 mission, a private astronaut mission lasting roughly 3.5 days in microgravity. Four private astronaut subjects participated in Pilot Egress Fitness from that mission. These subjects did not do any of the suited extravehicular activity (EVA) portion of the study, but had the opportunity to egress the SpaceX Dragon capsule itself shortly after landing, once the capsule was retrieved and placed on the recovery ship. They also were able to complete the capsule egress task using our standard mockup, a few hours after landing, allowing for direct comparison of simulated capsule egress to actual egress. All 4 subjects were successfully able to egress the SpaceX Dragon capsule shortly after landing, as well as complete the mockup capsule egress task. Egress time was impacted by recovery time, such that actual capsule egress from the Dragon was nearly doubled from baseline, whereas the mock-up capsule egress, which occurred several hours after landing, had the same impact on only one subject, with 3 subjects showing a smaller change in pre- to post-flight egress times. Outside of the Fram2 opportunity, no other results from Pilot Egress Fitness were added. CIPHER Egress Fitness has enrolled 8 total subjects, with 1 withdrawal, 4 subjects now complete, but with only 1 subject able to complete every test session due to various facility and schedule reasons. One subject is actively participating in postflight testing and the remaining 2 subjects have completed all pre-flight baseline data collection and will complete post-flight testing per the crew’s flight schedule. Notably, the 3 subjects who performed the post-flight EVA circuit on R+1 were able to complete all tasks, which is consistent with what was seen with the 4 Pilot Study subjects who were also able to complete all tasks. General performance and overall comfort of the astronauts clearly improves on R+4 as compared to R+1, but data analysis is still pending completion of all subjects. Beyond the changes to Pilot Egress Fitness to include private astronaut missions, there are no protocol changes to report on since the last NASA Task Book report in January 2025. Schedule Update: CIPHER Egress Fitness (CEF) has enrolled 8 total subjects, with 1 withdrawal. CEF01 has completed all pre-flight sessions, CEF02 completed the suit fit and L-180 EVA familiarization session, but we lost the L-60 EVA session due to an adverse incident investigation (resulting from another study) that shut down the NASA Active Response Gravity Offload System (ARGOS) from May 2023 with return to duty by late February 2024. CEF03 withdrew. CEF04 was capsule egress only, due to ARGOS shutdown. CEF05, CEF06, and CEF08 have completed all preflight testing. CEF07 completed all testing but we lost the R+1 EVA due to an issue with the ARGOS facility. Pilot Egress Fitness remains open in case of potentially scientifically interesting private astronaut missions. No subjects are currently enrolled.

Bibliography Type:	Description: (Last Updated: 01/08/2026)
Conference Materials (Downloadable)	Norcross JR, Schlotman TE, Cox LI, Rhodes R, Strock N, Young M, Marshall-Goebel KJ. "Validation of Fitness for Duty Standards Using Pre- and Postflight Capsule Egress and Suited Functional Tasks in Simulated Reduced Gravity." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025 Egress Fitness_HRPIWS_2025.pdf